

FactoryBench: Four-Level Causal Evaluation Hierarchy

Rigorous QA benchmark for LLM agents on causal machine understanding with industrial time-series data

Causal Evaluation Hierarchy

Strict dependency: success at Level N requires Level $N-1$

Increasing Causal Depth



L1 State Identification

Interpreting current machine states and detecting anomalies from sensor data.



L2 Intervention Reasoning

Predicting the immediate impact of control changes or faults at the present timestep.



L3 Counterfactual Analysis

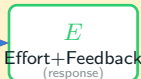
Simulating alternative historical timelines to assess hypothetical past conditions.



L4 Decision Support

Synthesizing multi-step recovery procedures using sensor fusion and technical manuals.

Physical Causal Schema



Healthy: $E = f(S)$ — Fault: empirical deviations from $f(S)$

Faults manifest as $\|E - f(S)\| > \epsilon$ under context C

Scalable Q&A Generation Engine

80 Expert Templates

Expert-authored across all 4 levels

Industrial Datasets

FactoryWave · AURSAD · Simulated Robotic
Real + simulated sensor streams

Dynamic Variable Sampling

Deterministic ground truth · Vendi Score validates diversity

Multi-Select

Ranking

Tensor Prediction

Open answers

Baseline Performance: Accuracy vs. Causal Depth

